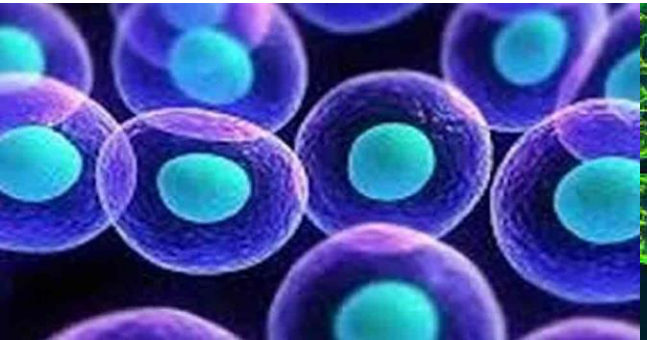
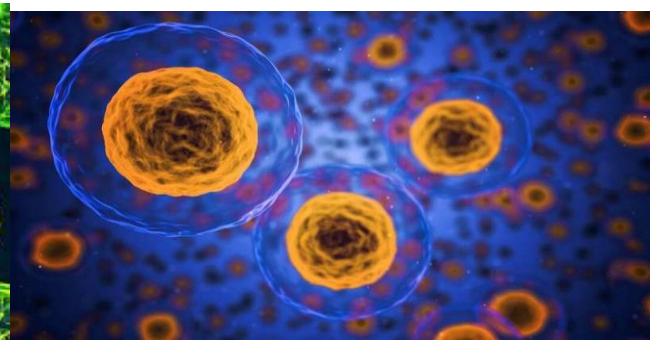
Part 9

Apoptosis, stem cell, and cancer





Apoptosis or programmed cell death/cellular suicide.

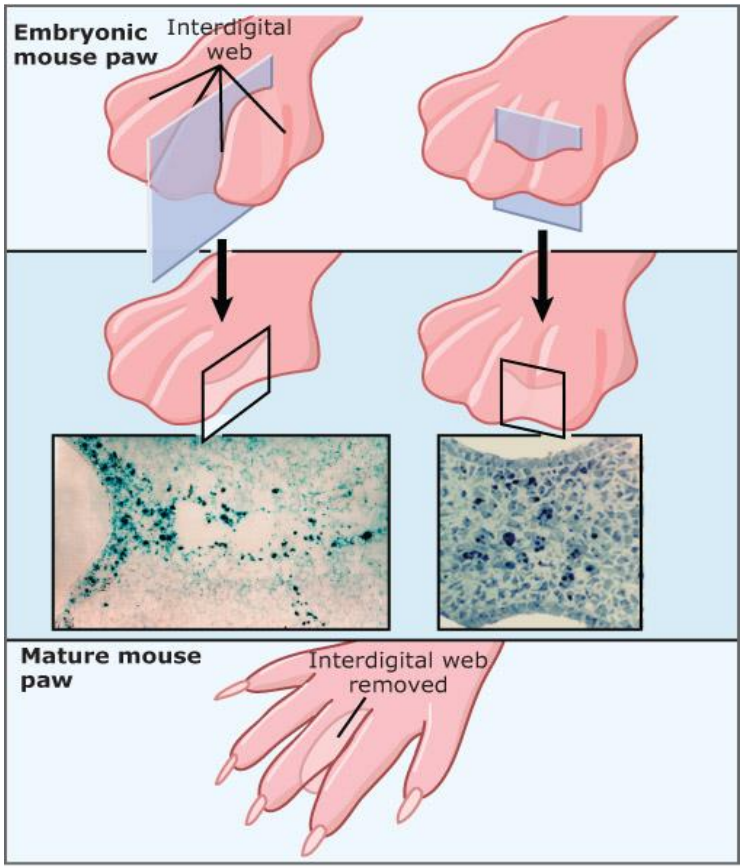
A general and convenient way to remove cells that should no longer be part of the organism

Some cells need to be “deleted” during development

Some cells are abnormal and could hurt the rest of the organism if they survive, such as cells with viral infections or DNA damage.

Cells in an adult organism may be eliminated to maintain balance – to make way for new cells or remove cells needed only for temporary tasks.

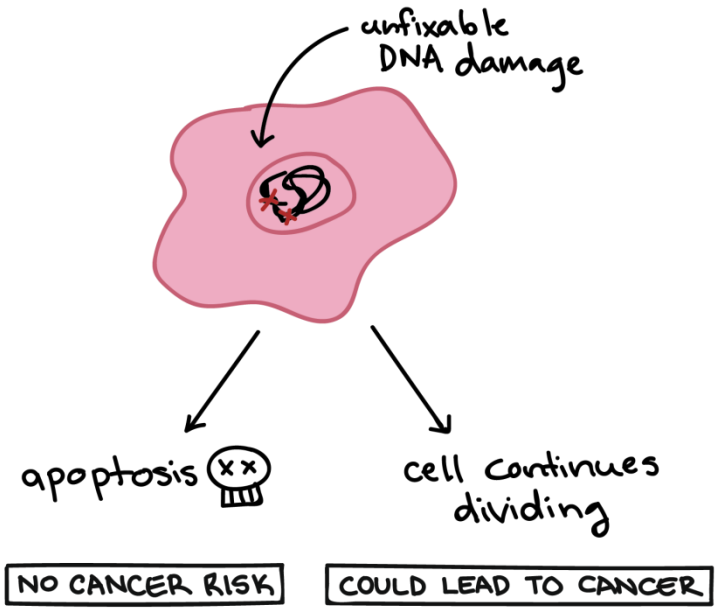
Programmed cell death is a normal part of development



This process occurs in all sorts of vertebrate species that have finger- or toe

Sections of the interdigital web show cell death (dark staining nuclei). This cell death has the characteristics of apoptosis

Apoptosis can eliminate infected or cancerous cells



Cells with DNA damage, pre-cancerous cells, and cells infected by viruses

→ Need to be removed undergo apoptosis

Apoptosis is key to immune function

Apoptosis also plays an essential role in the development and maintenance of a healthy immune system

* When B and T cells (immune cells that bind specific molecules) are first produced

→ They're tested to see if they react against any of the body's own "self" components

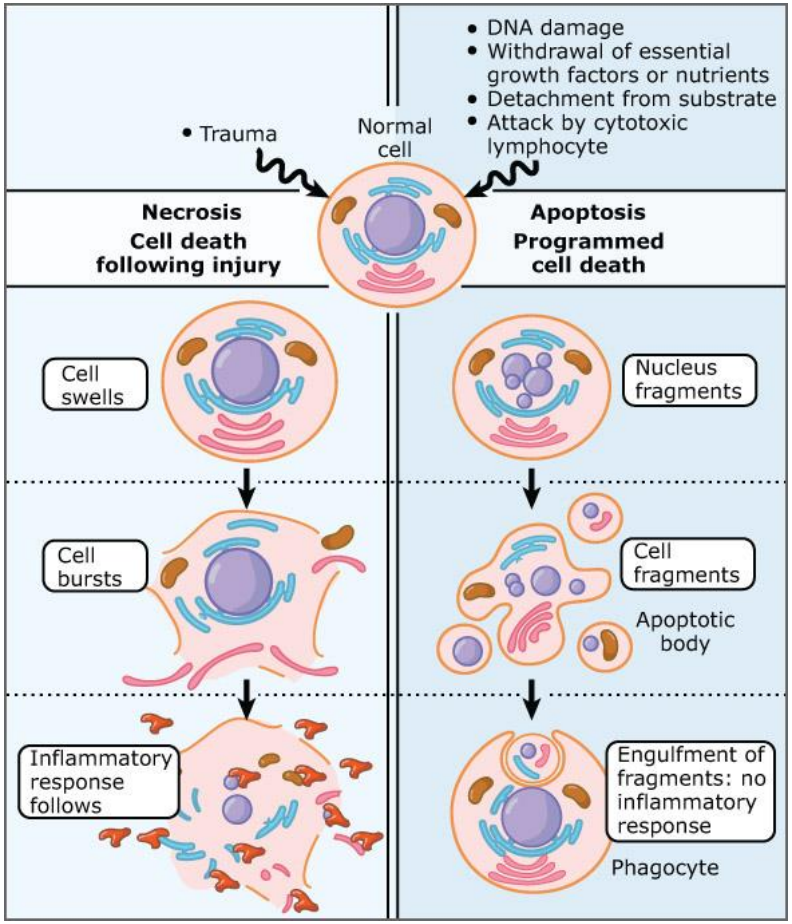
If yes → apoptosis

If apoptosis fail→ autoimmune conditions

* Pathogen is detected→ immune cells increase → pathogen is cleared → immune cells: apoptosis

Apoptosis vs Necrosis

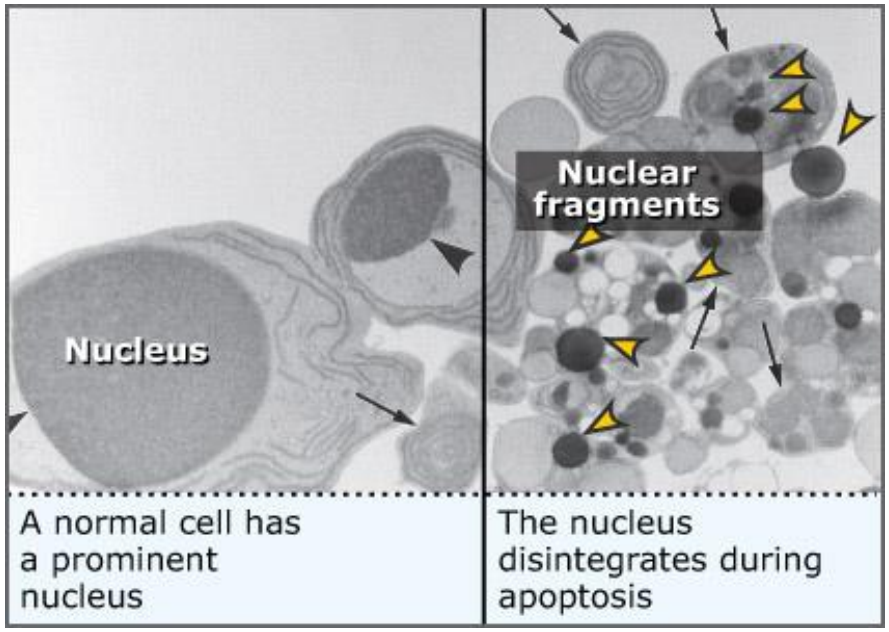
Cellular damage can result in necrosis, which has a different appearance than apoptosis, as organelles swell and the plasma membrane ruptures, without chromatin condensation



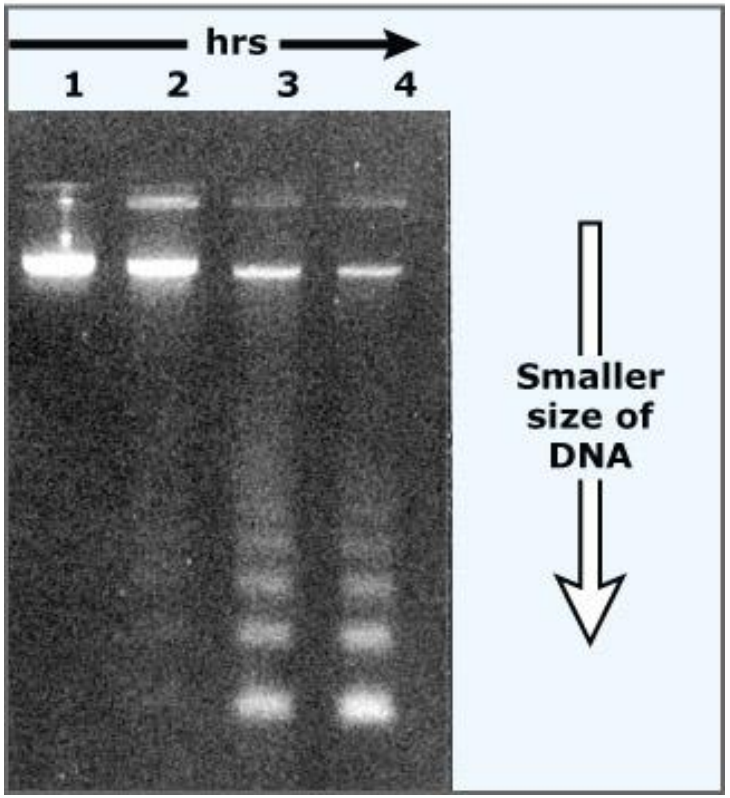
Apoptosis vs autolysis

Apoptosis: a cell kills itself

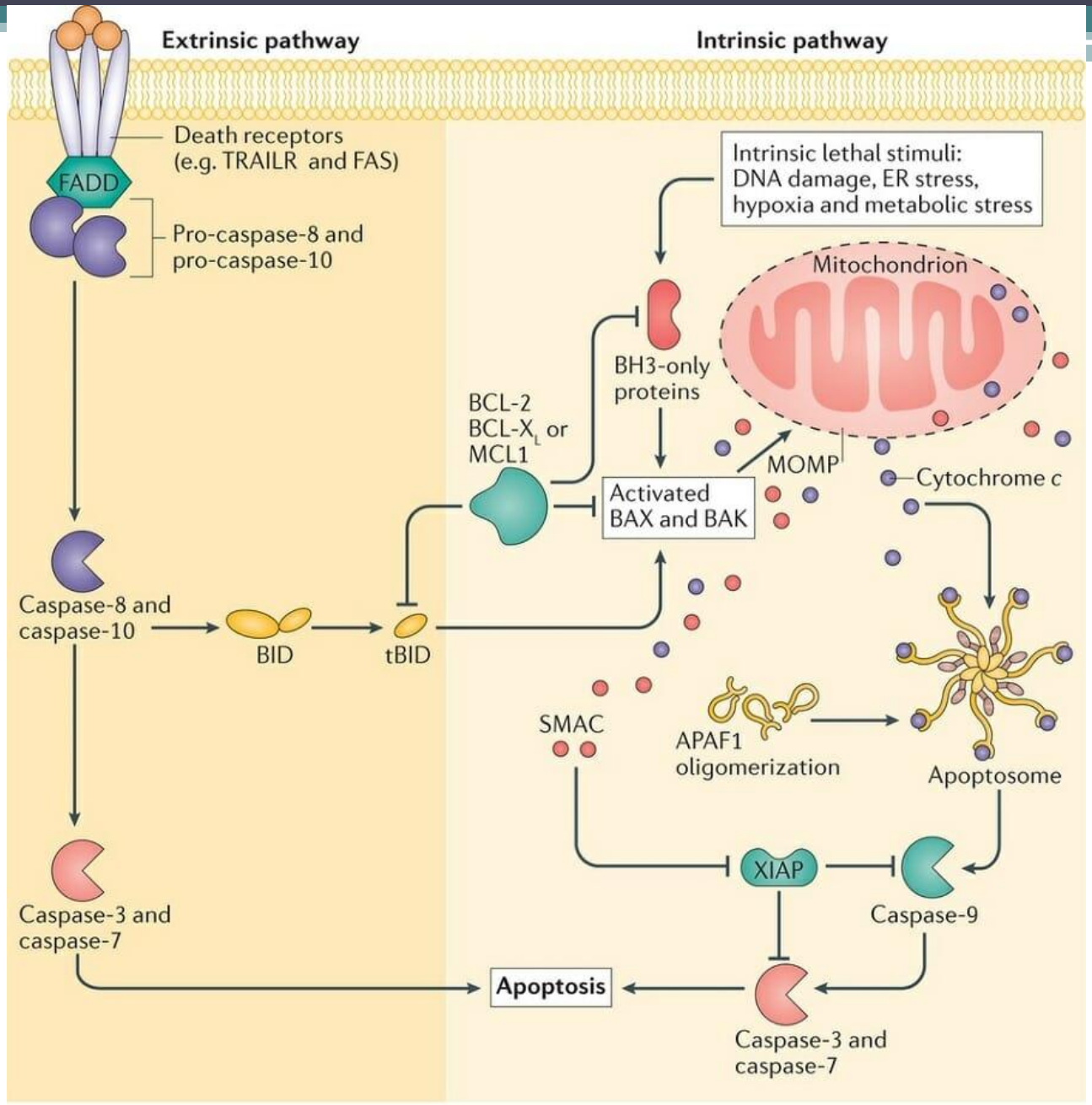
Autolysis: enzymes kill a cell



Cell structure changes during apoptosis. The left panel shows a normal cell. The right panel shows an apoptosing cell; gold arrows indicate condensed nuclear fragments



Fragmentation of DNA occurs in cells undergoing apoptosis.

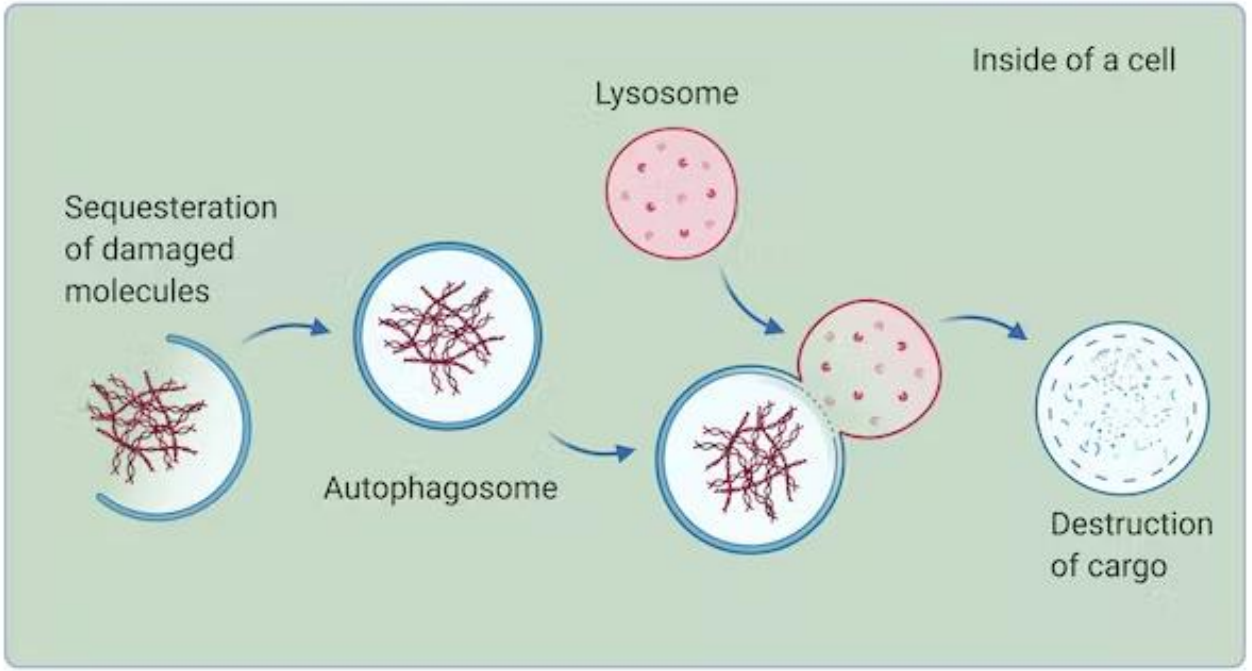

Extrinsic pathway
Intrinsic pathway
Death receptors
(e.g. TRAILR and FAS)
FADD
Pro-caspase-8 and
pro-caspase-10
Intrinsic lethal stimuli:
DNA damage, ER stress,
hypoxia and metabolic stress
Mitochondrion
BH3-only
proteins
BCL-2
BCL-X$_L$ or
MCL1
MOMP
Cytochrome c
Activated
BAX and BAK
Caspase-8 and
caspase-10
BID
tBID
SMAC
APAF1
oligomerization
Apoptosome
XIAP
Caspase-9
Caspase-3 and
caspase-7
Apoptosis
Caspase-3 and
caspase-7



Autophagy

Mechanism by which a cell “eats itself,” digesting molecules and organelles that are damaged or obsolete

The process of autophagy



The conversation.com

How to induce autophagy?

Fasting: Fasting means that you stop eating for a certain amount of time.

Calorie restriction: decreasing the number of energy units, or calories, your body consumes.

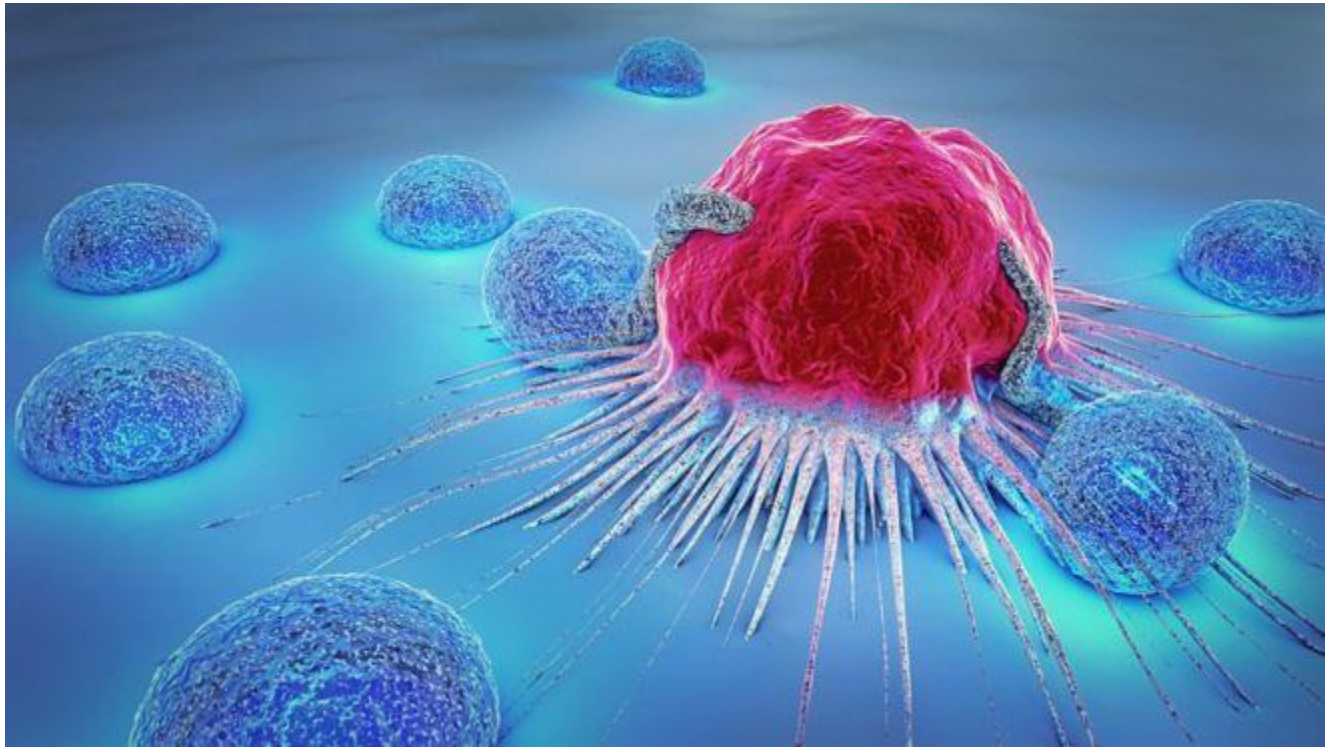
Switching to a high-fat, low-carb diet: This type of diet, commonly referred to as a keto diet

Exercise: Exercise stimulates processes that increase the activity of ATGs, such as stressing your skeletal muscles

Autophagy	Apoptosis
Balances the energy sources in the cell	Balance the number of cells in multicellular organism
Cause by cellular stress like starvation	Cause by intracellular programs
Lysosomes involved in the process	Lysosomes are not involved
Excessive leads to cell death	Excessive leads to atrophy
Allows the cell to survive stress	Does not allow the cell to survive

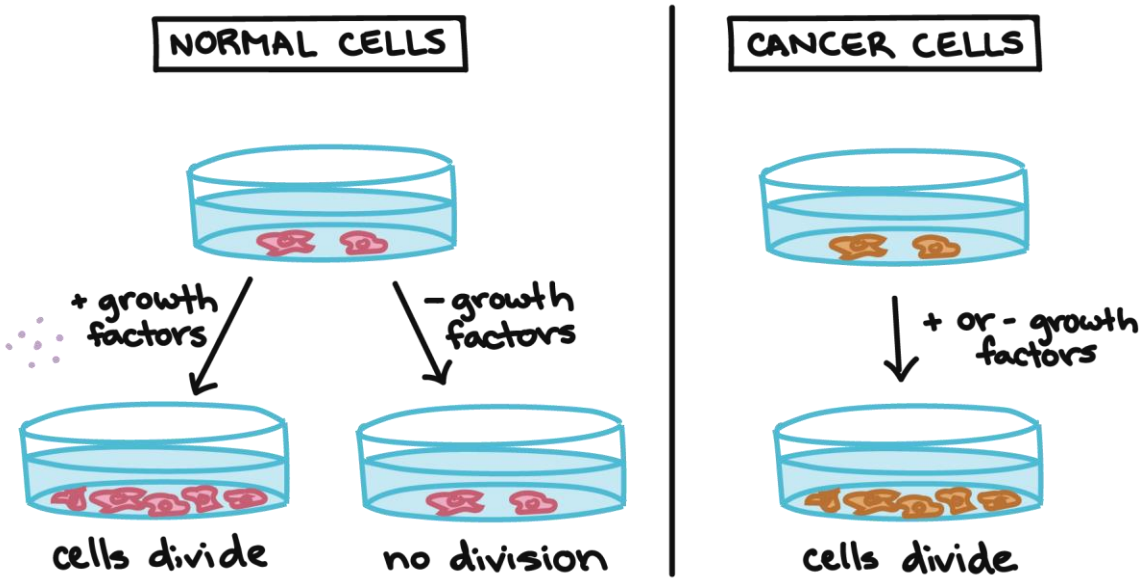
Cancer

Cancer is basically a disease of uncontrolled cell division



Cancer cells

* Multiply in culture without growth factors (outside of the body in a dish)



* Ignore signals that should cause them to stop dividing

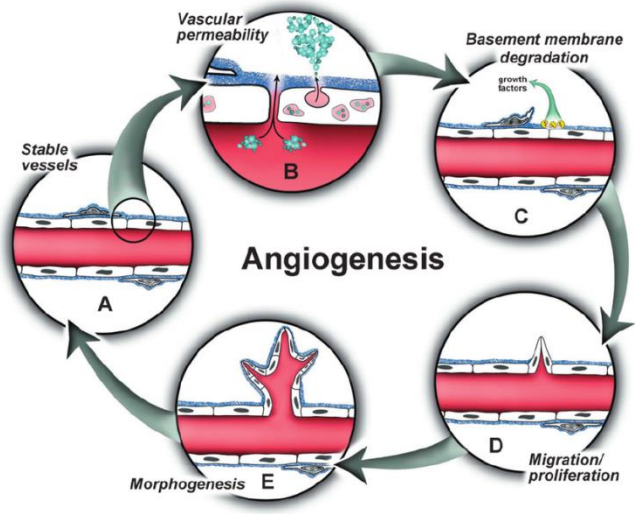
Cancer cells

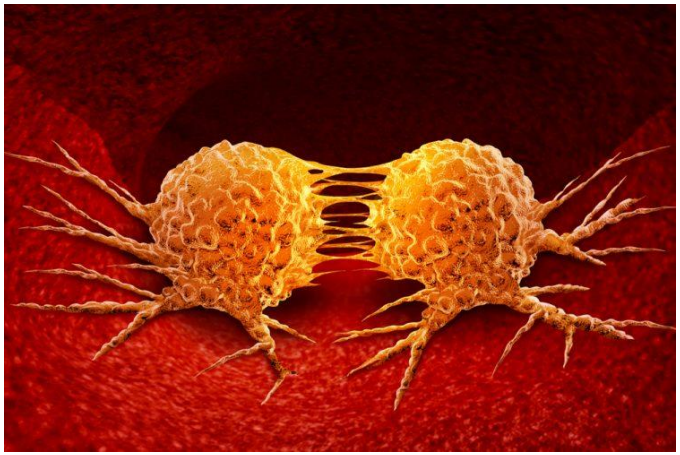
* Replicative immortality: can divide many more times than a normal cell

→ **telomerase**

* Promote growth of new blood vessels, a process called **angiogenesis** (which gives tumor cells a source of oxygen and nutrients)

* Gain the ability to migrate to other parts of the body →**metastasis**

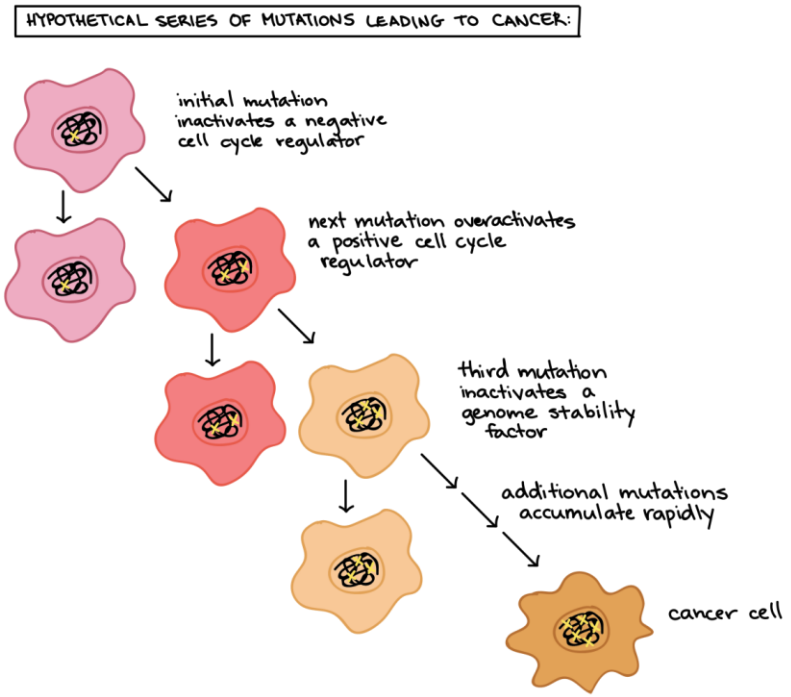


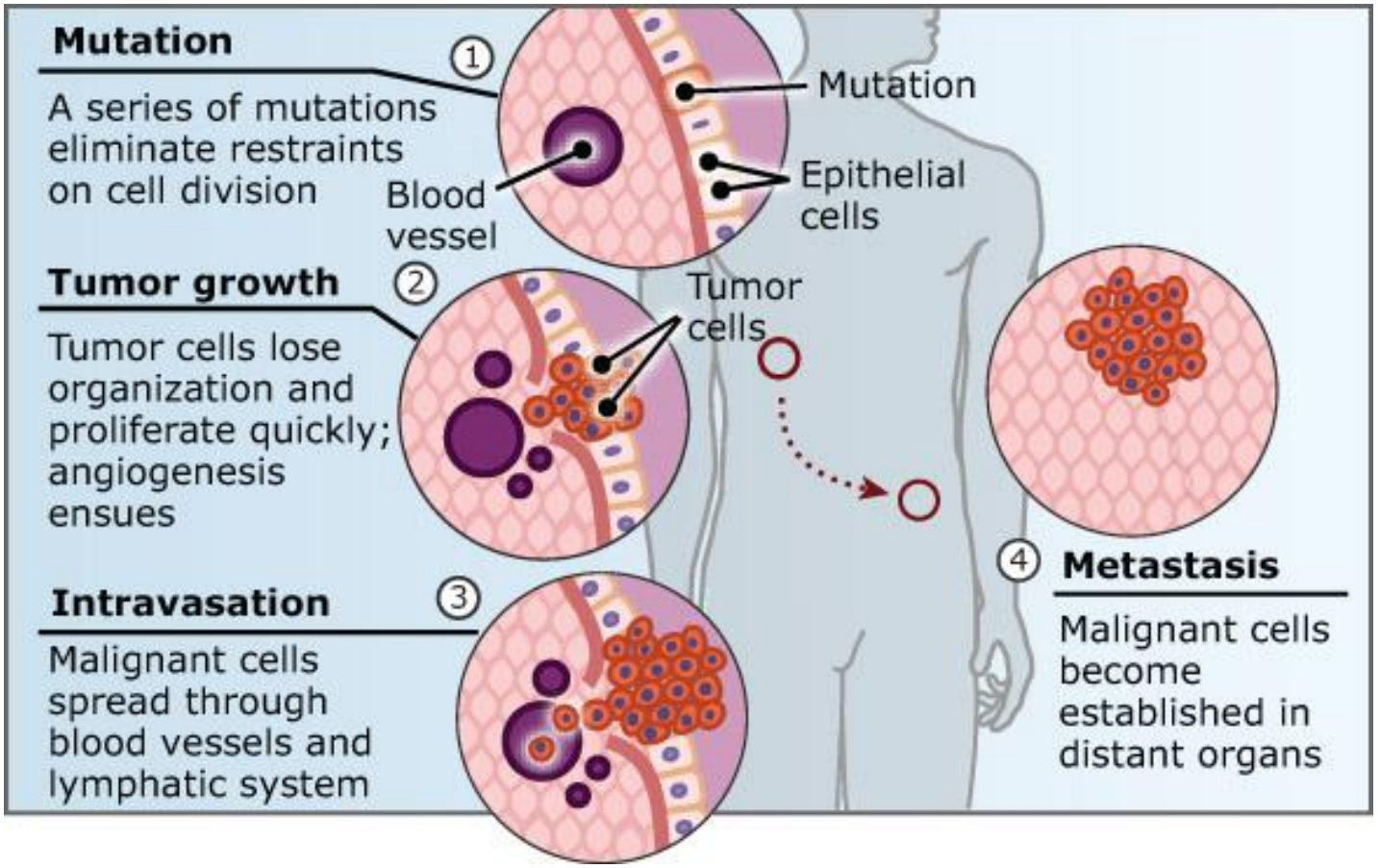


How cancer develops

Cancers develop by an accumulation of mutations

- Cancer is **fundamentally a genetic disease**: it arises as a consequence of pathological changes in the information carried by DNA
- ≠ others: somatic cells vs gem cells



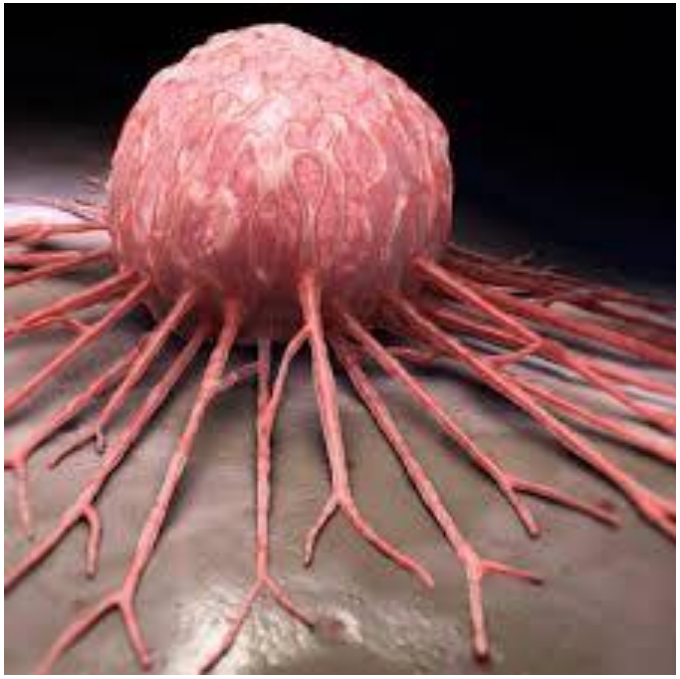


Cancer progresses from a local tumor to widespread metastasis.

Tumor

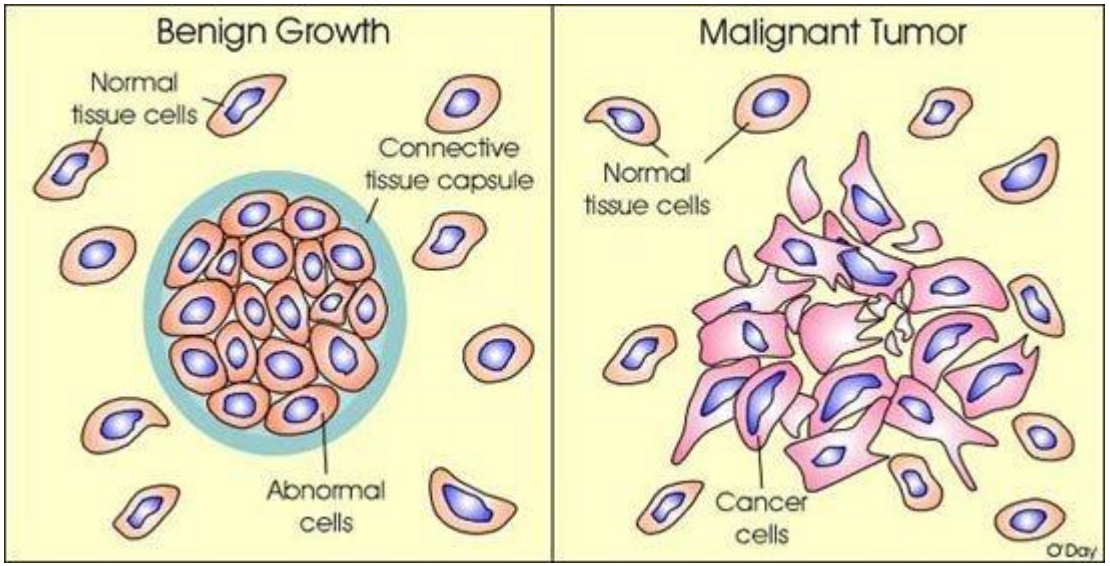
An abnormal mass of tissue resulting from uncontrolled division

- Malignant tumor (cancerous)
- Benign tumor (non-cancerous)



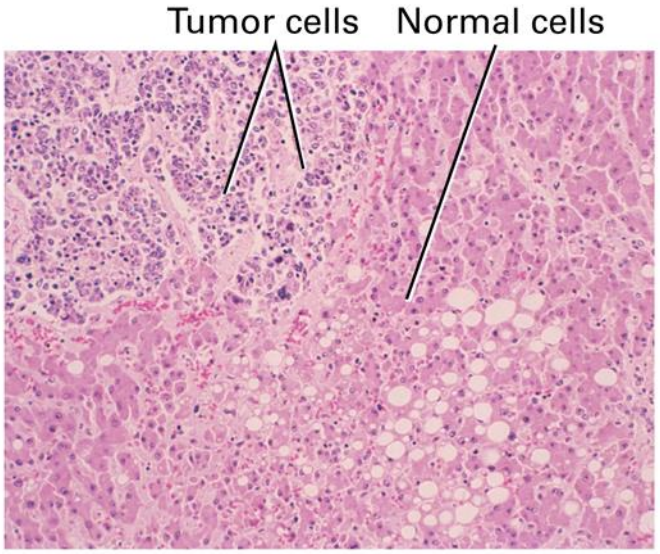
Benign tumor

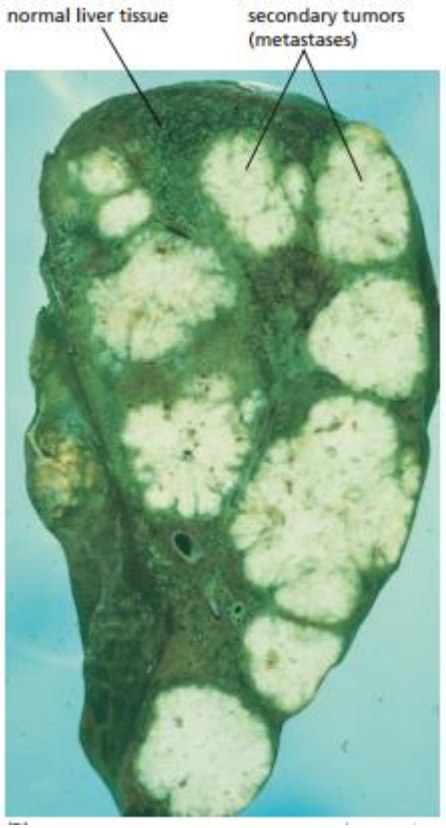
- Do not spread to other part of the body
- Generally localized and of small size
- Slow rate of division
- Cells that closely resemble, and may function, like normal cells
- Do not break out of originating organ
- Easily to removed by surgery



Malignant tumor

- Invade or spread to other parts of the body
- High rate of division
- Spread by forming metastasis
- Cells travel through circulation
- Very difficult to treat





Secondary tumors in a human liver, originating from a primary tumor in the colon.

Site	New Cases	Deaths
All sites	1,660,290	580,350
Oral cavity & pharynx	41,380	7,890
Digestive system	290,200	144,570
Respiratory system	246,210	163,890
Soft tissue (including heart)	11,410	4,390
Bones and joints	3,010	1,440
Skin (excluding basal & squamous)	82,770	12,650
Breast	234,580	40,030
Genital system	339,810	58,480
Urinary system	140,430	29,790
Eye & orbit	2,800	320
Brain & other nervous system	23,130	14,080
Endocrine system	62,710	2,770
Lymphoma	79,030	20,200
Multiple myeloma	22,350	10,710
Leukemia	48,610	23,720
Other & unspecified primary sites*	31,860	45,420

* More deaths than cases suggests lack of specificity in recording underlying causes of death on death certificates.

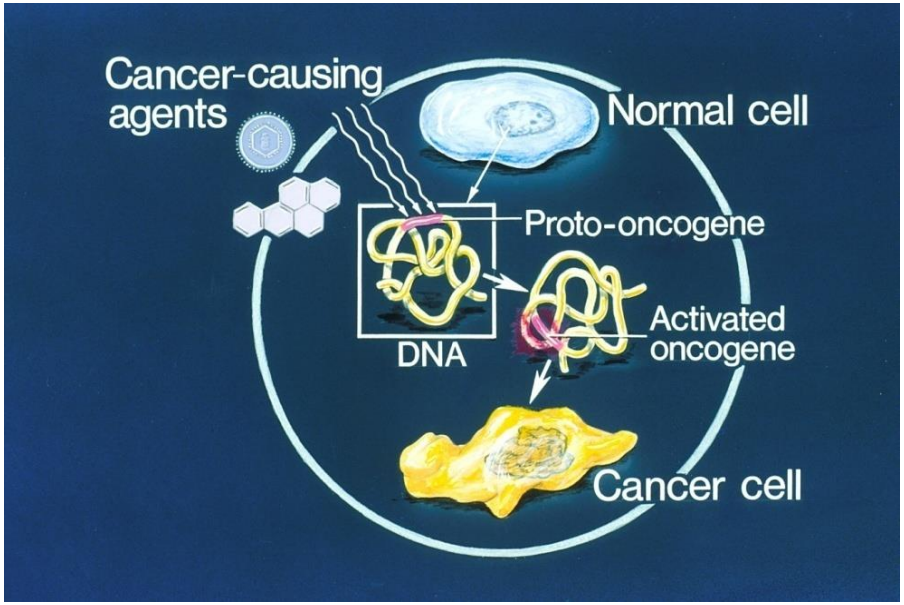
The frequency certain tissue types become cancerous, the course of disease development, and efficacy of treatment is variable.

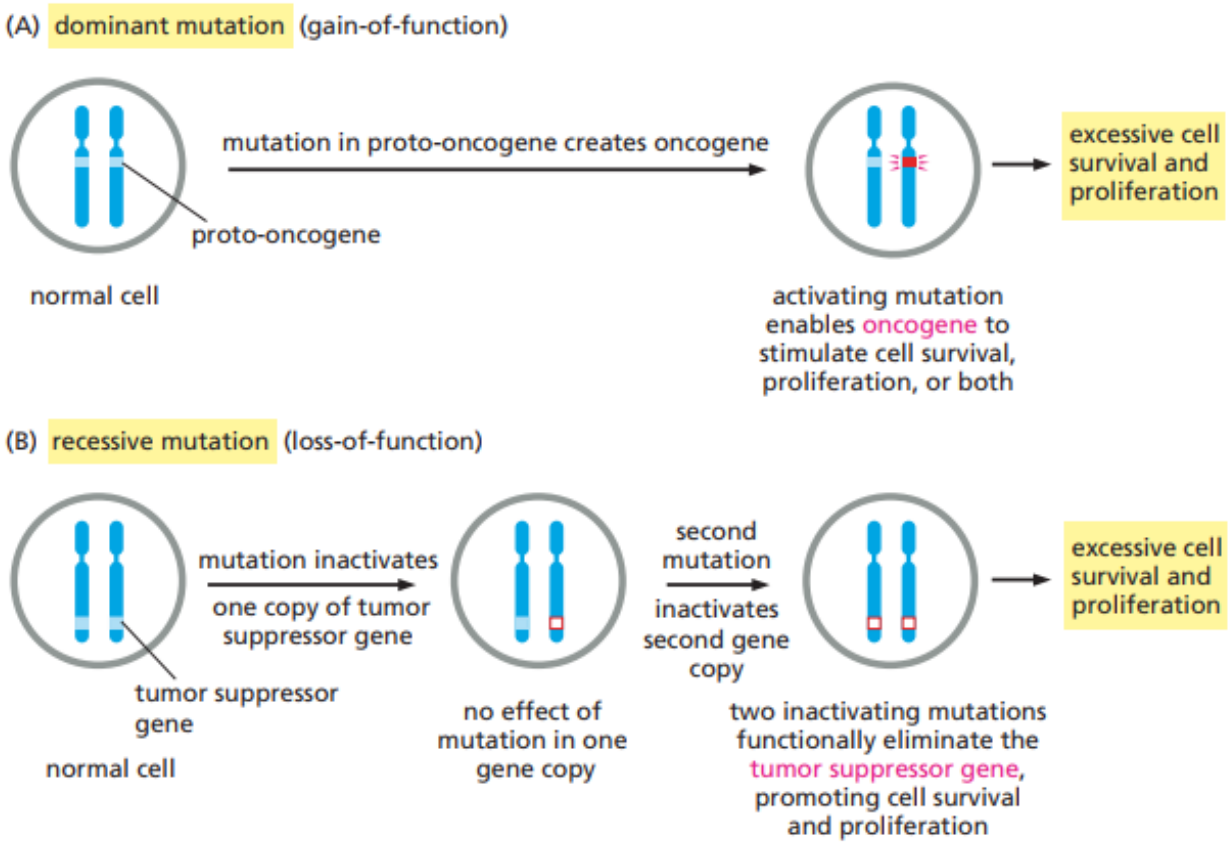
Cell cycle regulators and cancer

* Two types of cell cycle regulators may promote the development of cancer

+ Positive regulators may be over activated (become **oncogenic**)

+ Negative regulators (called **tumor suppressors**) may be inactivated





Genes that are critical for cancer are classified as protooncogenes or tumor suppressor genes, according to whether the dangerous mutations are dominant or recessive.

(A) Oncogenes act in a dominant manner: a gain-of-function mutation in a single copy of the proto-oncogene can drive a cell toward cancer.

(B) Loss-of-function mutations in tumor suppressor genes generally act in a recessive manner: the function of both copies of the gene must be lost to drive a cell toward cancer

Positive regulators : over-activated

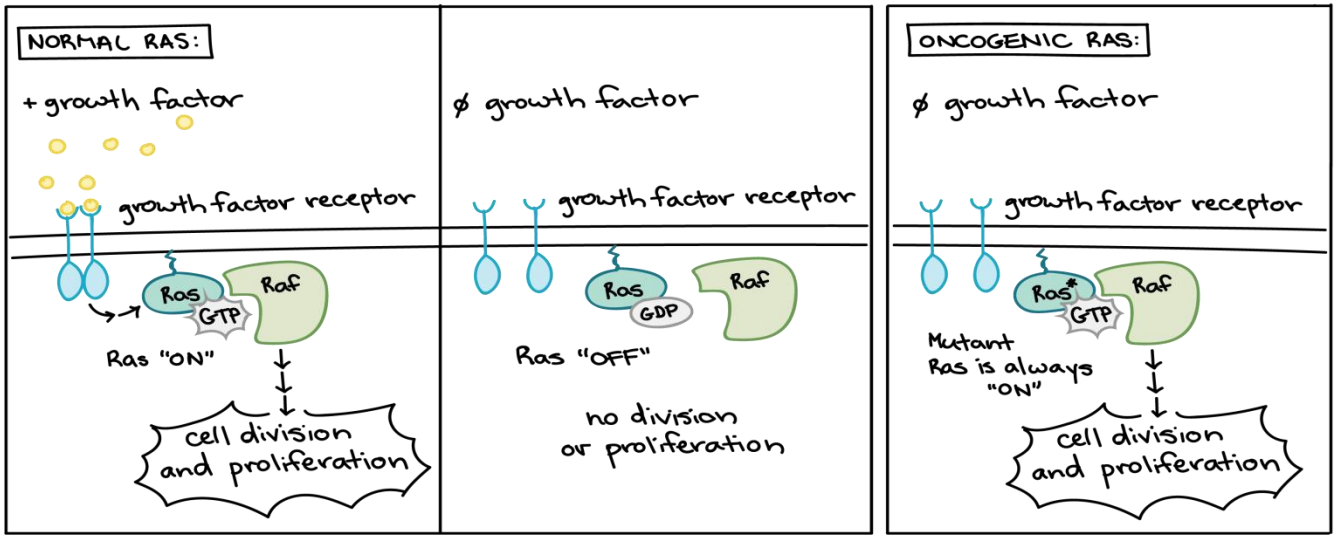
A growth factor receptor may send signals even when growth factors are not there→ over-activated

The overactive forms of these genes are called **oncogenes**, while the normal, not-yet-mutated forms are called **proto-oncogenes**

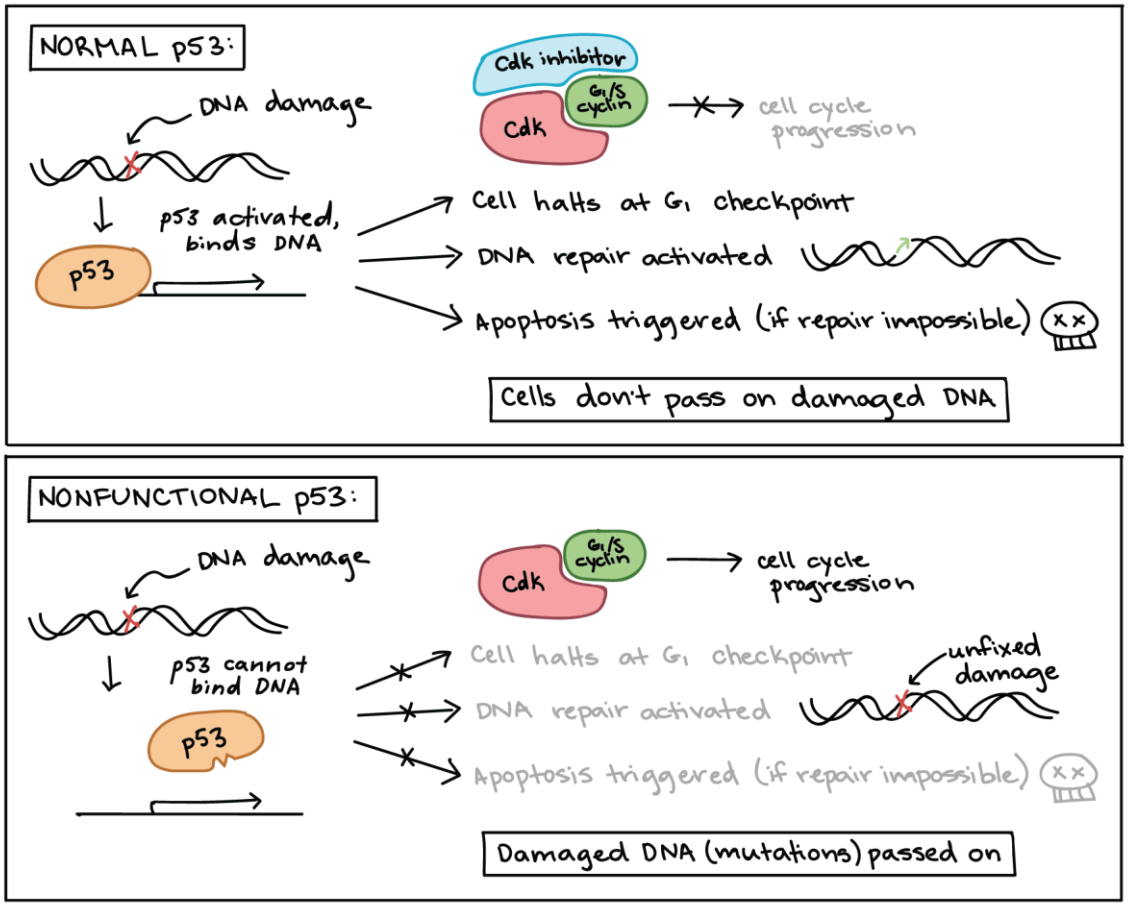
Proto-oncogenes —mutation→ oncogenes

↓

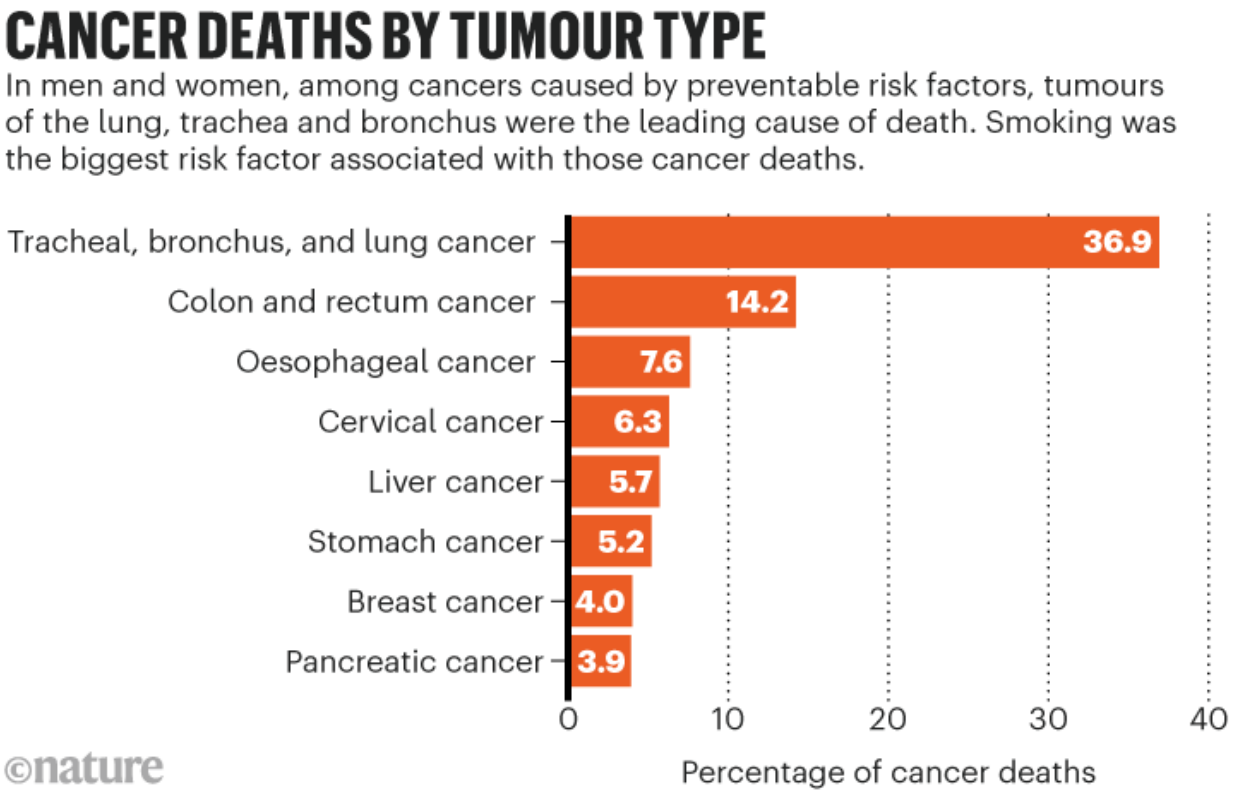
cancer



Tumor suppressors



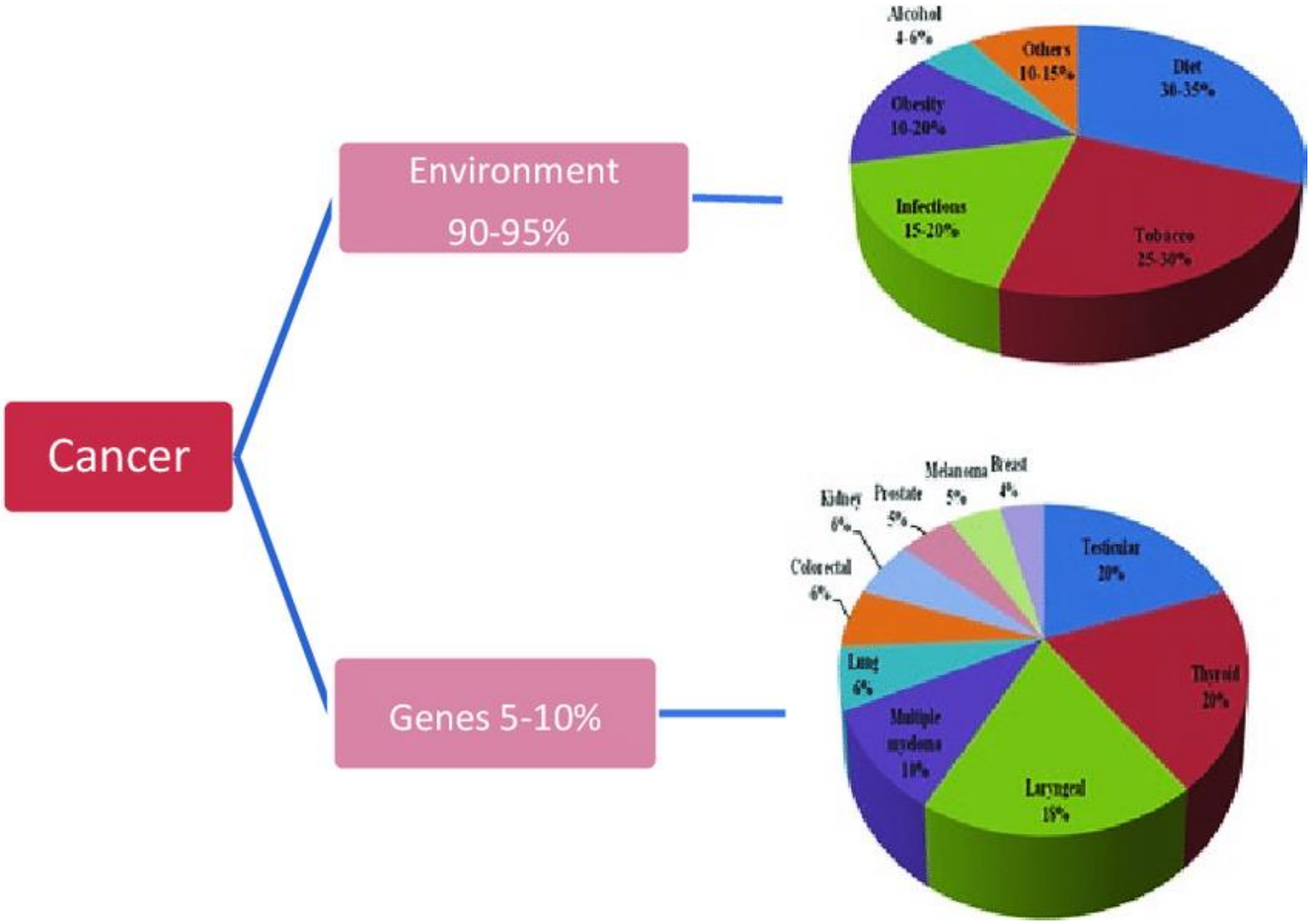
Epidemiological studies identify Preventable causes of cancer



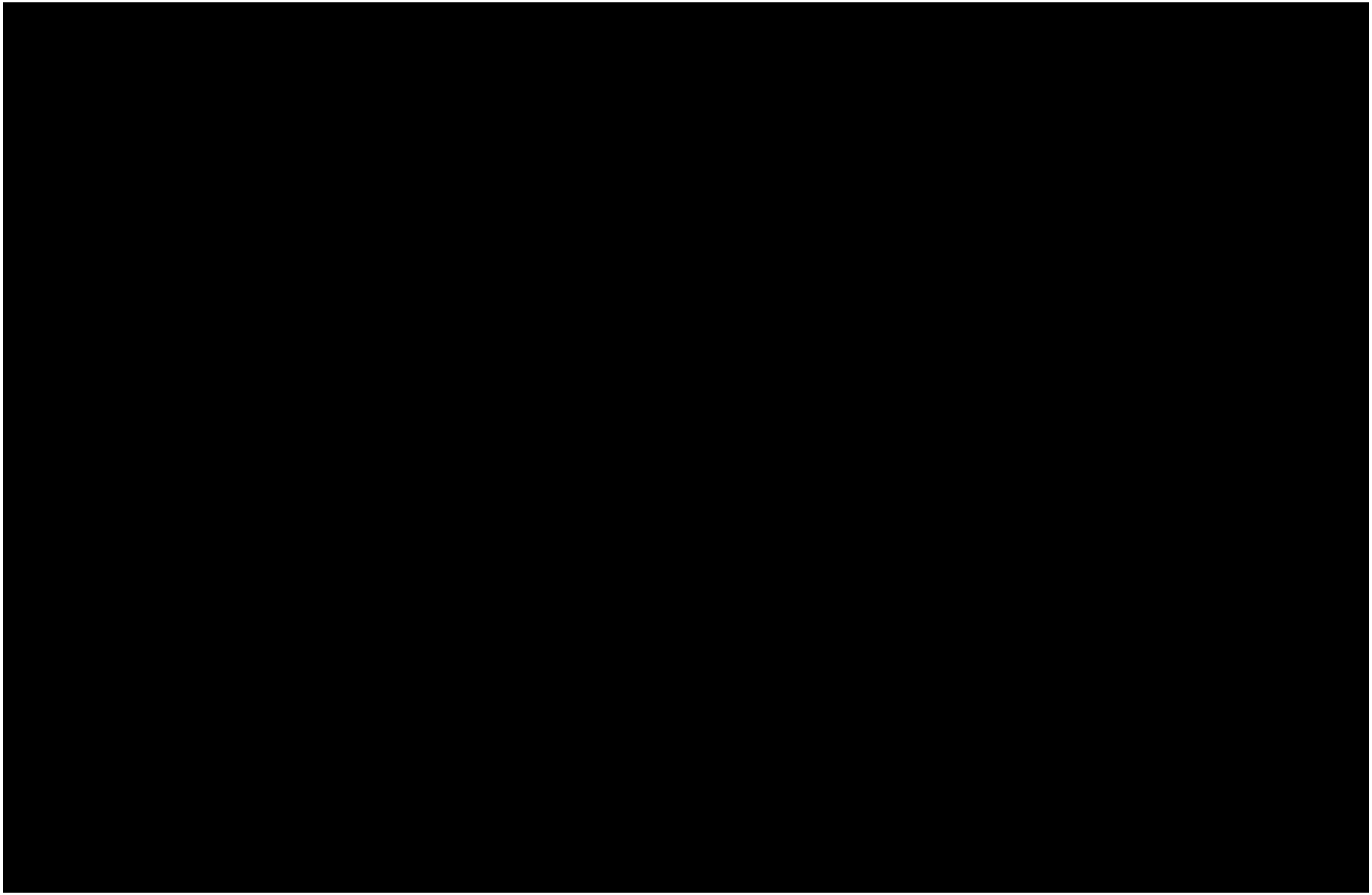
doi: https://doi.org/10.1038/d41586-022-02355-x

Nearly 50% of cancer deaths worldwide are caused by preventable risk factors, such as smoking and drinking alcohol, according to the largest study of the link between cancer burden and risk factors1.

Epidemiological studies identify Preventable causes of cancer

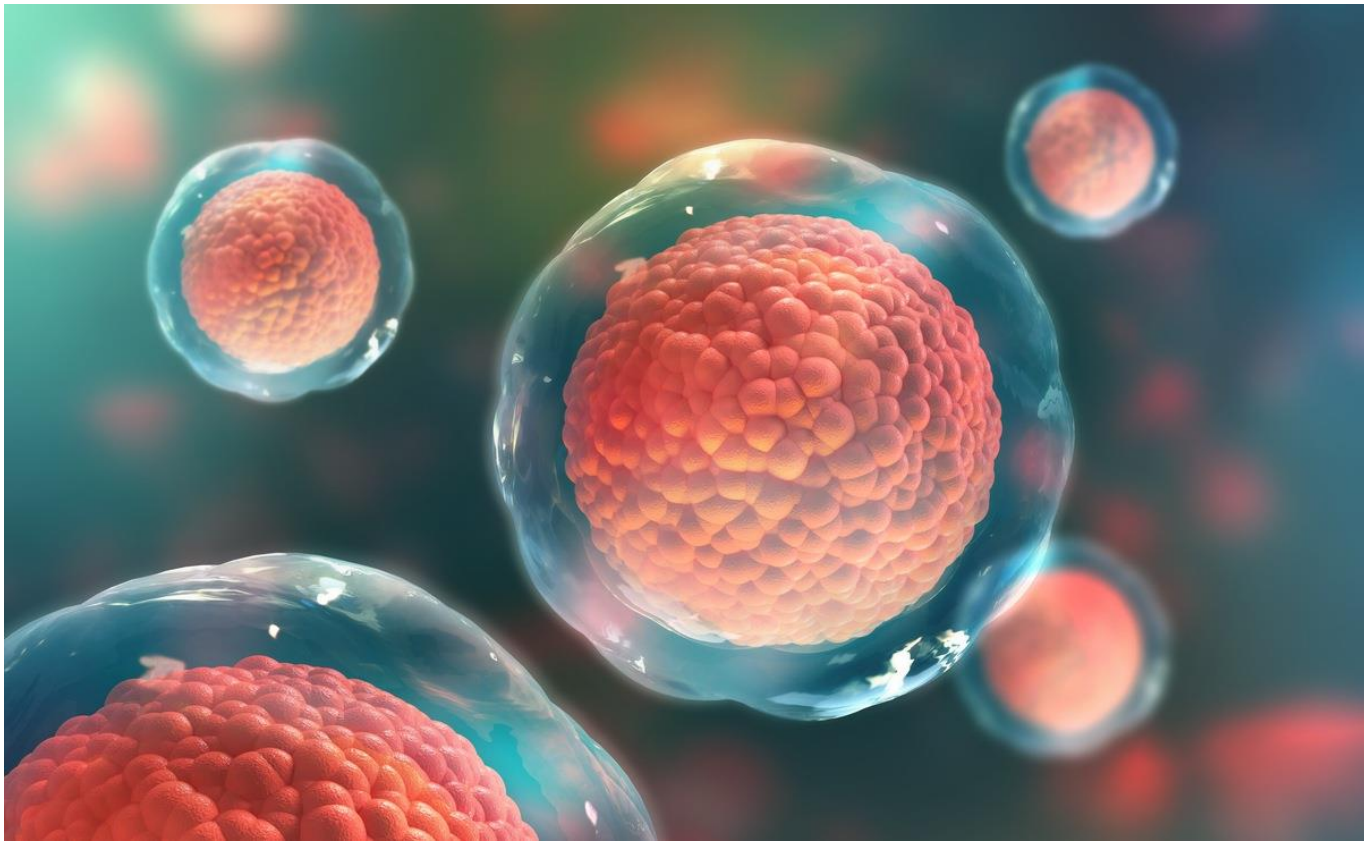


DOI:10.1007/978-981-13-7607-8_1



Stem Cells

Cells with the potential to develop into many different types of **cells** in the body

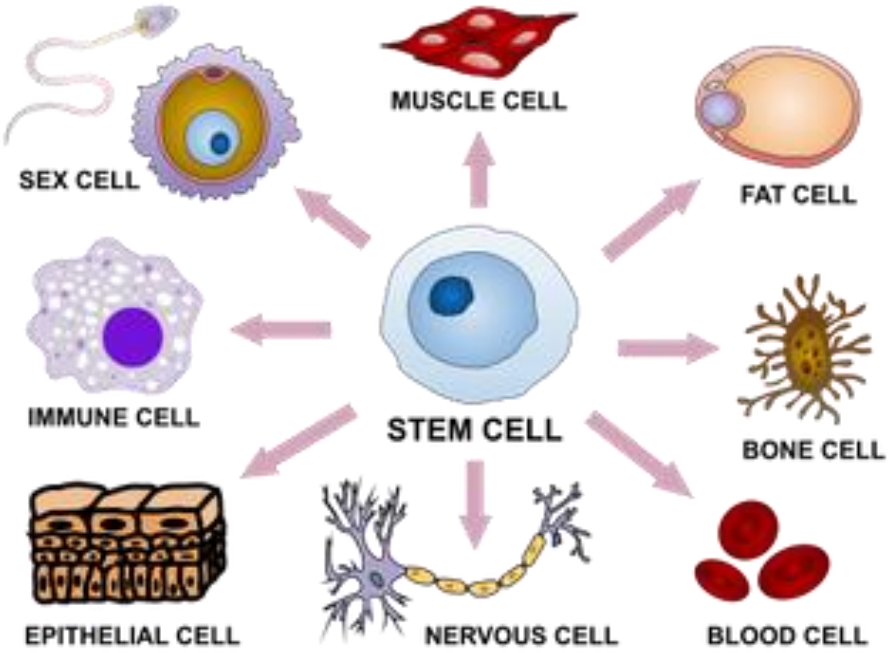


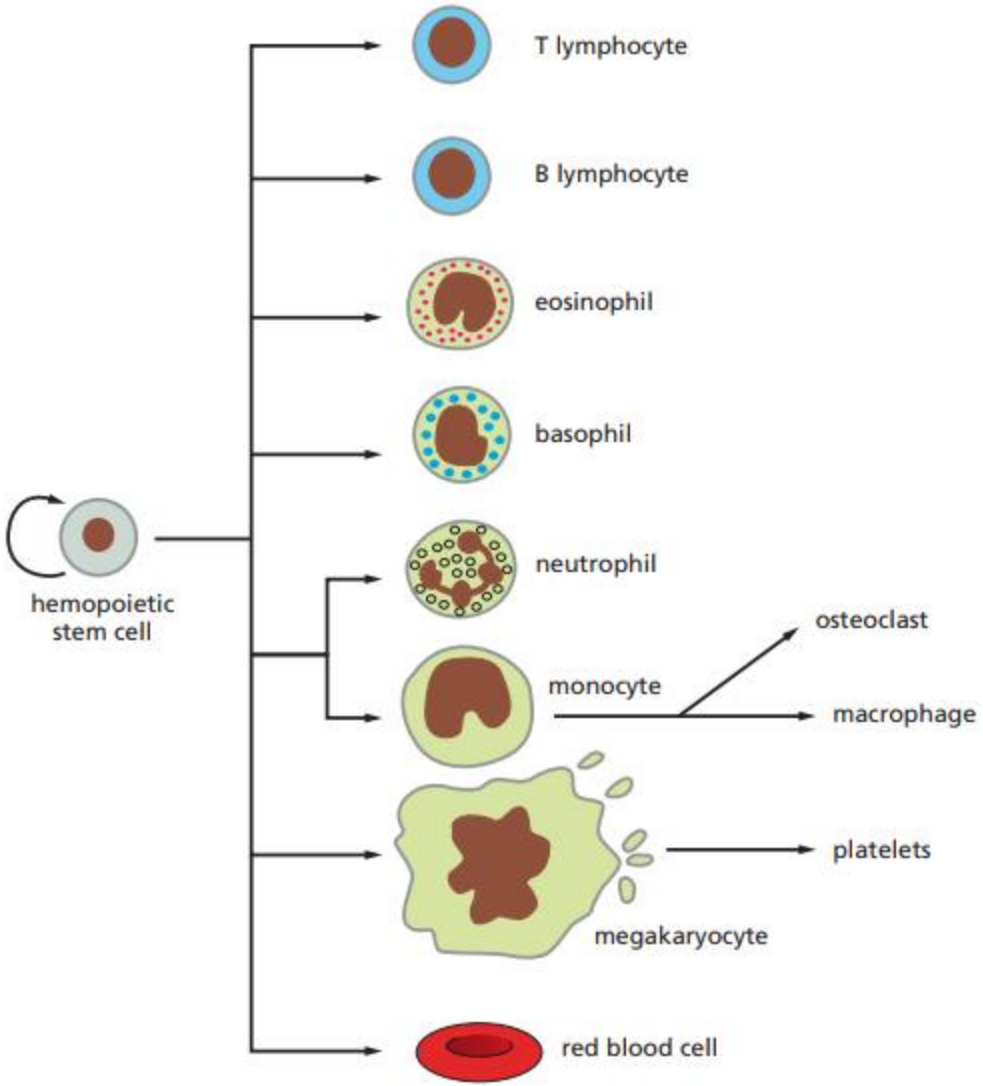
“Blank” cells/ Unspecialized

Proliferation: Renewing themselves for long periods of time

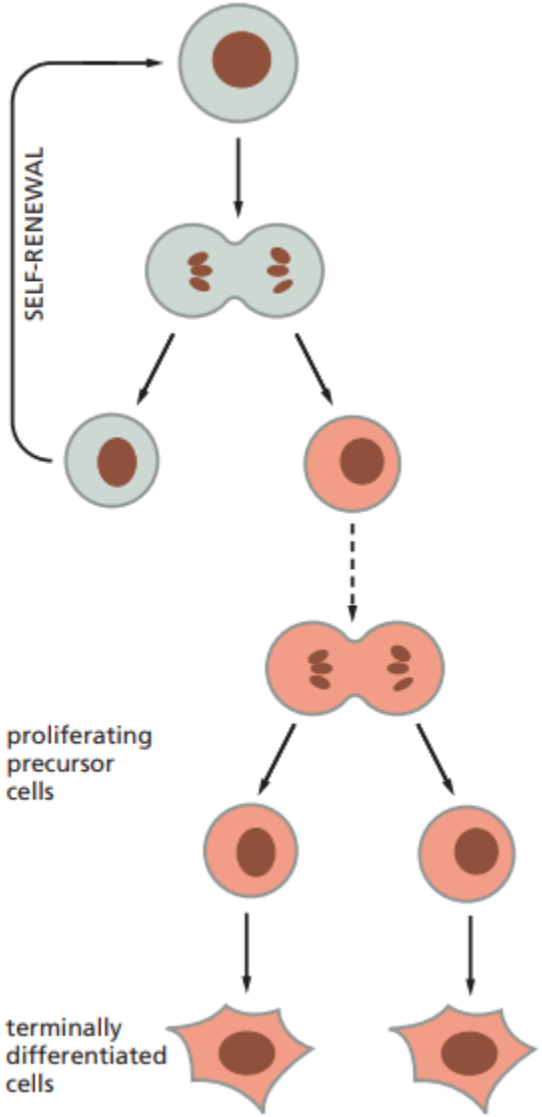
Differentiation: Have the potential to give rise to specialized cells types

Plasticity



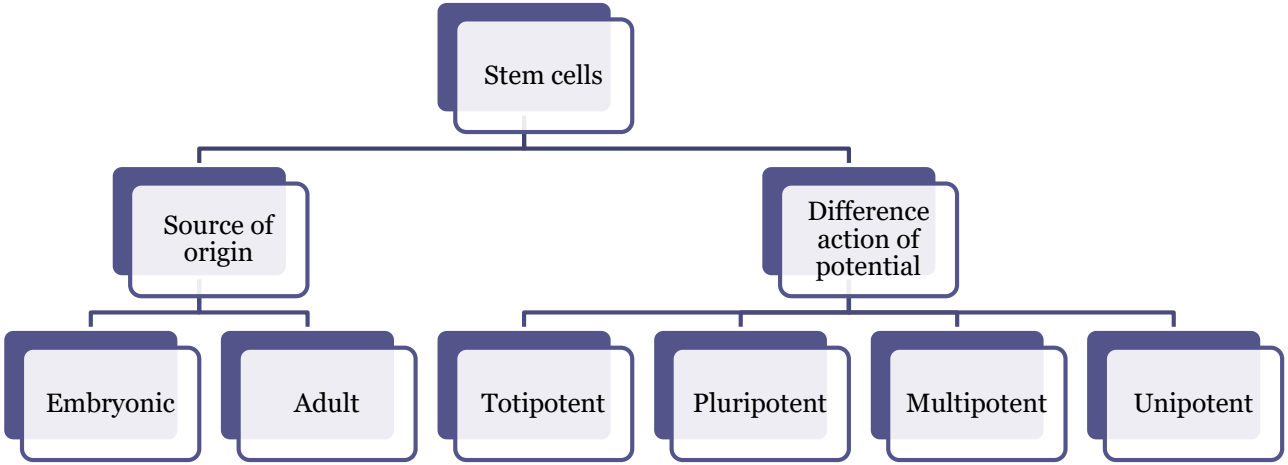


A hemopoietic stem cell divides to generate more stem cells, and differentiate into the mature blood cell types

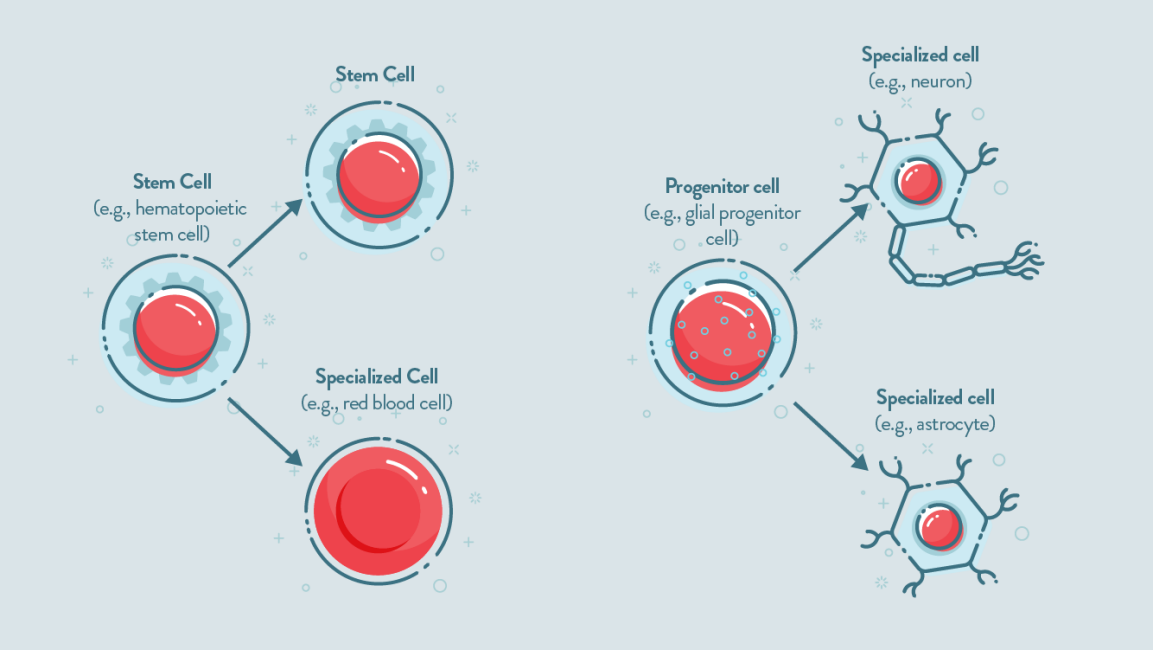


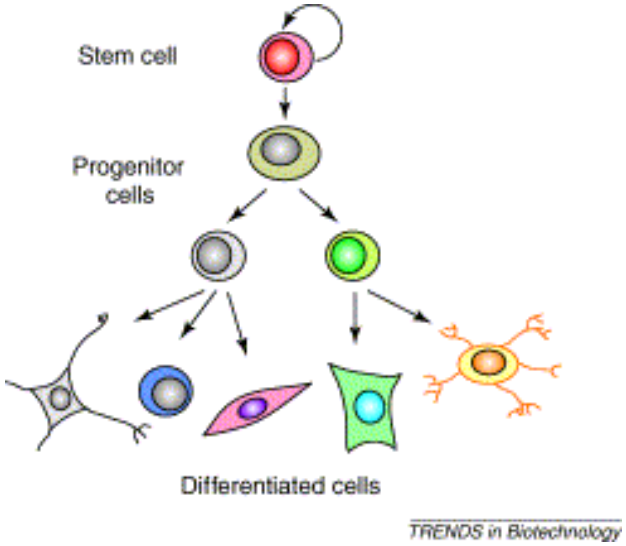
When a stem cell divides, each daughter can either remain a stem cell or go on to become terminally differentiated.

The terminally differentiated cells usually develop from precursor cells that divide a limited number of times before they terminally differentiate.


Stem cells
Source of origin
Difference action of potential
Embryonic
Adult
Totipotent
Pluripotent
Multipotent
Unipotent

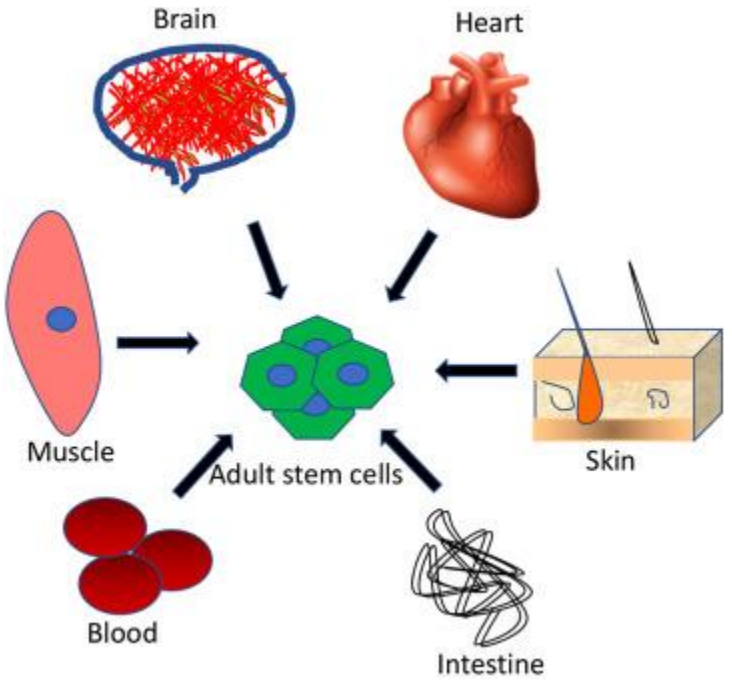
Stem cells divide slowly





Adult stem cells

Found in small numbers in most adult tissues (bone marrow, fat…)



Source

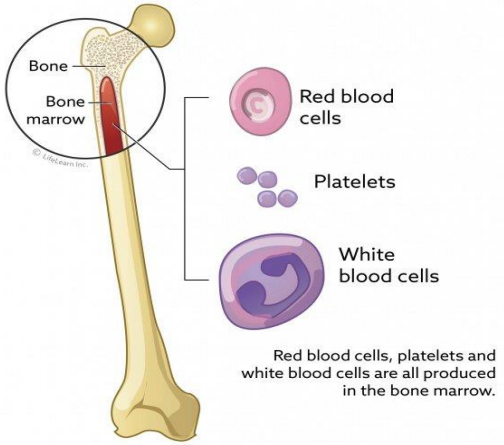
Umbilical cord

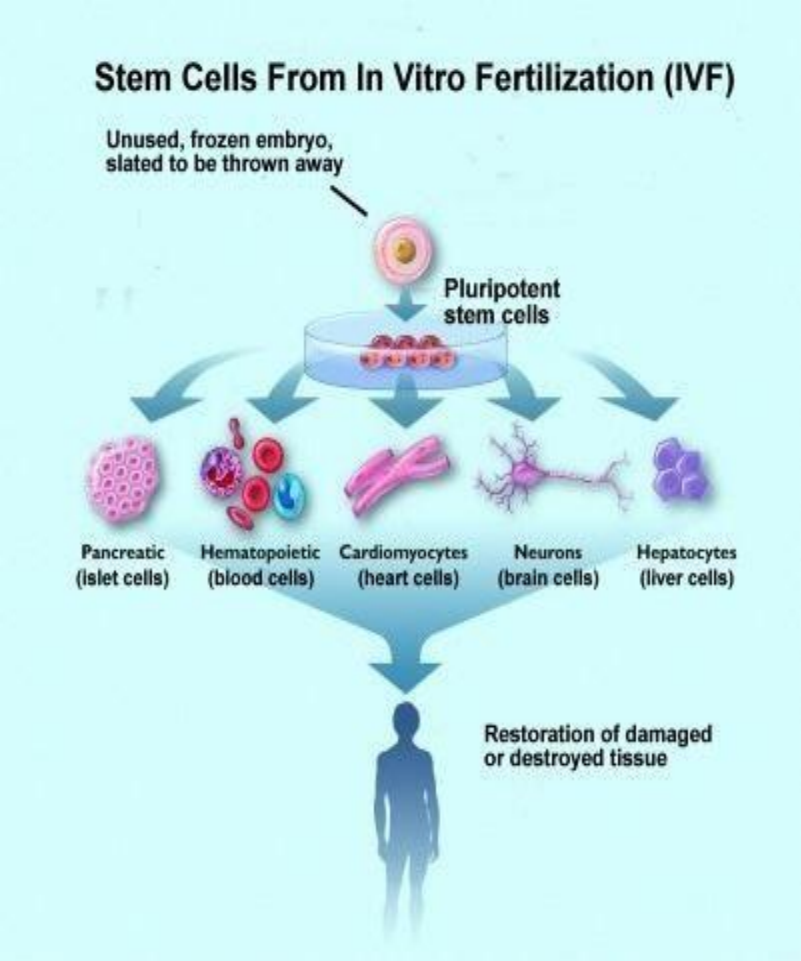


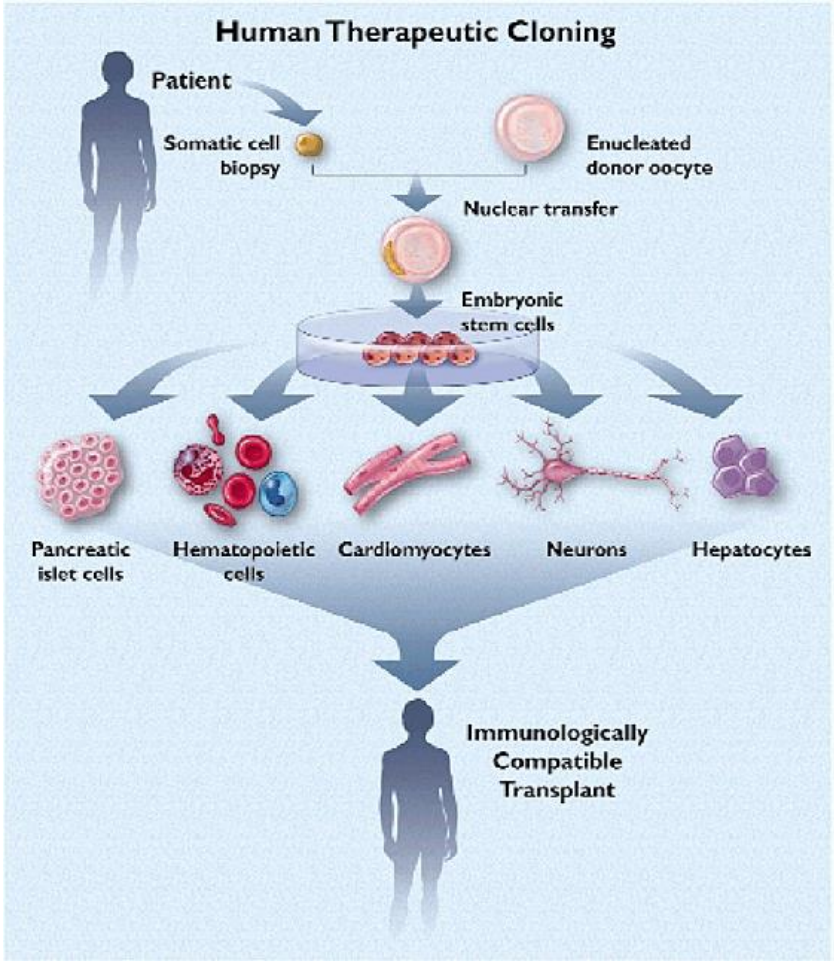
Cord blood cells



Body







Somatic Cell Nuclear Transfer (SCNT)

Embryonic

- Totipotent

- Large numbers can be harvested

- May cause immune rejection

- Function to generate new organs and tissues

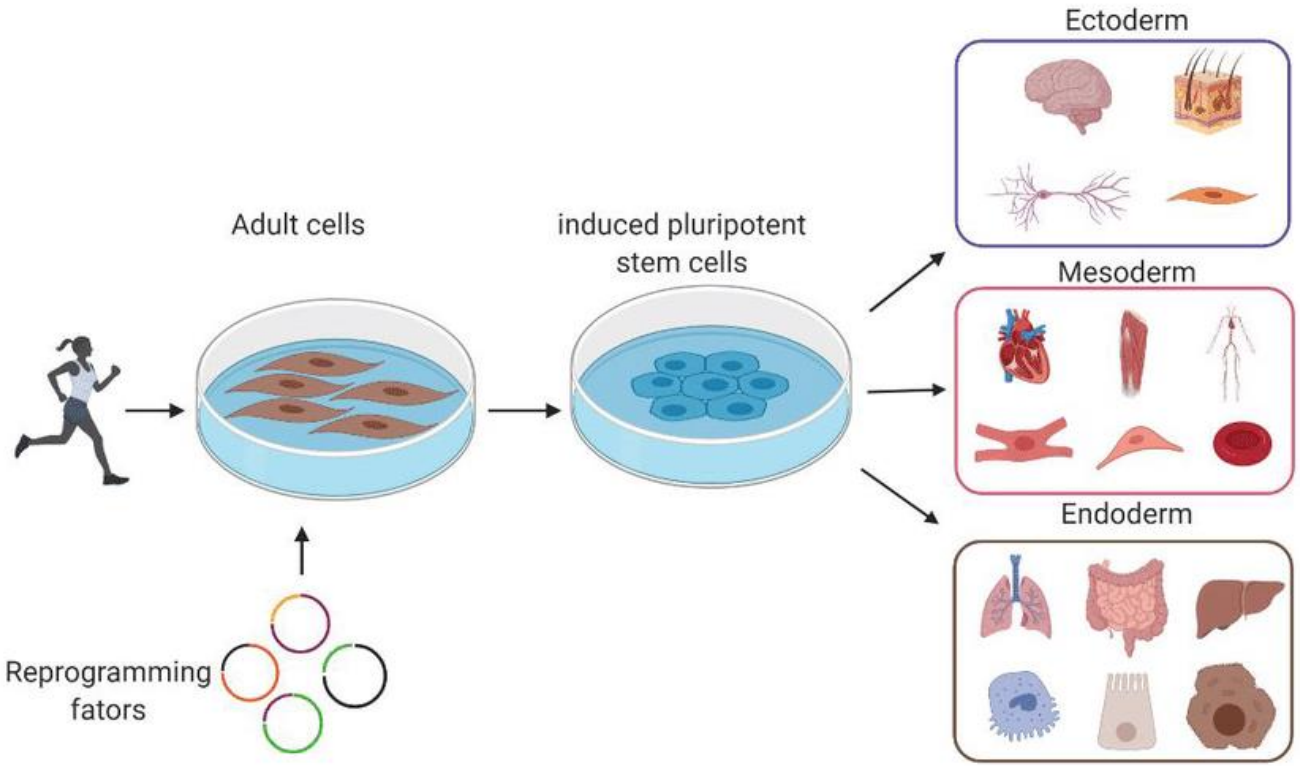
Adult stem cells

- Multi or pluripotent

- Limited numbers, difficult to isolate

- Less likely to cause immune rejection, since the patient's own cells can be used

- Function the replace cells during the natural course of cell turnover

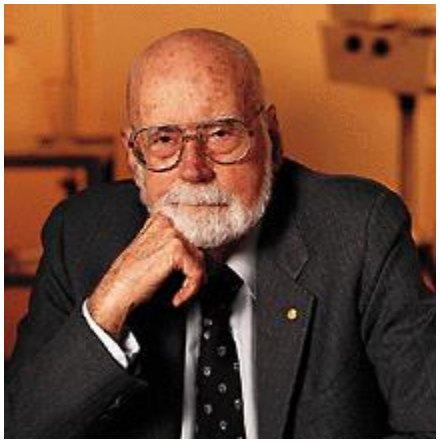


Induced pluripotent stem cells are reprogrammed adult human cells from patients or healthy individuals. Once reprogrammed to an early development stage, induced pluripotent stem cells can self-renew and be differentiated into all cell types of the human body.

Purpose: Cell of variable potency that can self-renew

Normal: used for organ maintenance or repair

Therapeutic: Repair damaged organ by introducing/injecting extra stem cells into
Work for hematopoietic system, skin cells…



The Nobel Prize, 1990

E. Donnall Thomas (1920-2012)

First successful hematopoietic stem cell transplantation (HSCT) in treatment of leukemia